Stress assignment without morphology in Portuguese: the qualityless vowel /0/ and the Metrically Marked Morpheme

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Abstract:

This paper offers an Optimality-Theoretic reinterpretation and synthesis of the author's earlier Japanese works — Makino (2010, 2012) — on Portuguese stress patterns, focusing exclusively on the theoretical essentials. We put forward a **prosody-only account of Portuguese word stress**. The system assumes a **qualityless vowel /0/ carrying one mora** in underlying representations, employed by specific morphemes. The default stress pattern arises from right-edge-aligned, iterative binary trochaic footing from the final syllable of the prosodic word — governed by the ranking *CLASH, #WSP ≫ PARSE(σ) ≫ FTBIN; TROCHEE; ALIGN-R(Ft, PrWd); ALIGN-R(HdFt, PrWd). On the other hand, a class of ***metrically marked morphemes*** — **MMMs** — triggers footing from the penultimate syllable only when their own final vowel serves as the nucleus of the penultimate syllable of the prosodic word, formalised as ALIGN-R(Ft, σ_{final}_MMM) ≫ ALIGN-R(Ft, PrWd).

Metrification is computed strictly from phonological information — including syllabification with Ç and syllable weight — together with a single piece of non-categorial lexical information, namely the MMM condition. No reference to syntactic categories or to morphological bracketing is made at any point in the stress assignment process. This approach therefore eliminates morphology from stress computation proper, attributes apparent exceptions to /0/ and MMMs, and unifies items such as *contador*, *contadora*, *falava*, *falávamos*, *faláveis*, *número*, *numeroso*, *numérico*, *sobre*, *após* under one minimal mechanism.

0. Abbreviations

The abbreviations, symbols and diacritics used in this paper are as follows:

PrWd = prosodic word; **Ft** = foot; **HdFt** = head foot; **MMM** = *metrically marked morpheme*; **σ_{final}_MMM** = the syllable whose nucleus is the final vowel of an MMM; **L** = light syllable; **H** = heavy syllable

/ / = underlying representation; **-** = morpheme boundary; **/0/** = qualityless vowel with one mora of quantity; → = derivation; **.** = syllable boundary; **C̩** (e.g. r̩, n̩, s̩) = consonantal syllablic nucleus that arises from the association of a consonant with the stray/floating mora contributed by /0/; **( )** = footing; ˈ = primary stress; ˌ = secondary stress; **[ ]** = surface representation; **#** = word boundary; **σ** = syllable

FTBIN = foot binarity; **TROCHEE** = trochaic foot; **ALIGN-R(Ft, PrWd)** = align the right edge of each foot with the prosodic word; **ALIGN-R(Ft, σ_{final}_MMM)** = align the right edge of the foot with the syllable whose nucleus is the final vowel of an MMM; **ALIGN-R(HdFt, PrWd)** = align the right edge of the head foot with prosodic word; **PARSE(σ)** = parse every syllable into a foot; ***CLASH** = avoid a stress clash; **#WSP** = weight to stress principle

1. The qualityless vowel /0/

A **qualityless** vowel **/0/** that carries **one mora** in underlying representations is assumed — e.g. *contador* /kont-a-dor-**0**/ (cf. *contadora* /kont-a-dor-**a**/); *jacaré* /ʒakarɛ-**0**/; *trabalhar* /trabaʎ-a-r**0**/; *trabalharmos* /trabaʎ-a-r**0**-mos/; *após* /a-pɔs**0**/; *até* /atɛ**0**/. This vowel is a purely phonological entity, serving various morphological purposes: in *contador* /kont-a-dor-**0**/, the /0/ functions as the thematic vowel — the stem vowel or *índice temático* (stem marker) — just as /a/ does in *contadora* /kont-a-dor-**a**/. On the other hand, it forms part of the infinitive or future subjunctive suffix /-r0/ in *trabalhar* /trabaʎ-a-r**0**/, and likewise occurs in the preposition *até* /atɛ0/, which does not display any internal structure.

2. Postulation of the MMM

It is also assumed that there exists a classe of ***metrically marked morphemes*** **(MMMs)** that are lexically specified to require **iterative binary footing from the penultimate syllable of the PrWd**, but **only when their own final vowel serves as the nucleus of that penultimate syllable** — e.g. *-ic-* /-ik-/ in *acadé**mico*** /akadɛm-**ik**-o/; *-vel* /-vel/ in *agradá**vel*** /a-grad-a-**vel**-0/; *numer-*

/numɛr-/ in *número* **/numɛr**-o/; *-va-* /-va-/ in *trabalhá**va**mos* /trabaʎ-a-**va**-mos/; *-r-* /-r0-/ in *trabalha**r**des* /trabaʎ-a-**r0**-des/. **Whether a morpheme belongs to the phonological class of MMMs is independent of its morphosyntactic category**.

3. Unmarked metrification

Regardless of syntactic category or morphological structure, **iterative formation of trochaic binary feet proceeds from the final syllable of the PrWd towards the word-initial edge**, with the **rightmost foot** serving as the head and bearing **primary stress**:

possibilidade /pɔs-i-bil-i-dad-e/ → pɔ.si.bi.li.da.de → (ˌpɔ.si)(ˌbi.li)(da.de) → [ˌpu.siˌβɨ.liˈðaː.ðɨ]
contador /kont-a-dor-0/ → kon.ta.do.ɽ̥ → (ˌkon.ta)(ˈdo.ɽ̥) → [ˌkõ.tɐˈðoːɾ]
contadora /kont-a-dor-a/ → kon.ta.do.ra → (ˌkon.ta)(ˈdo.ra) → [ˌkõ.tɐˈðoː.ɾɐ]
jacaré /ʒakarɛ-0/ → ʒa.ka.rɛ.ɛ → (ˌʒa.ka)(ˈrɛ.ɛ) → [ˌʒɐ.kɐˈɾɛː]
coração /korason-0/ → ko.ra.so.n̥ → (ˌko.ra)(ˈso.n̥) → [ˌku.ɾɐˈsɐ̃ṵ̃]
trabalhamos /trabaʎ-a-mos/ → tra.ba.ʎa.mos → (ˌtra.ba)(ˈʎa.mos) → [ˌtrɐ.βɐˈʎɐː.muʃ]
trabalhais /trabaʎ-a-des/ → tra.ba.ʎa.des → (ˌtra.ba)(ˈʎa.des) → [ˌtrɐ.βɐˈʎai̯ʃ]
trabalhar /trabaʎ-a-r0/ → tra.ba.ʎa.ɽ̥ → (ˌtra.ba)(ˈʎa.ɽ̥) → [ˌtrɐ.βɐˈʎaːɾ]
sobre /sobre/ → so.bre → (ˈso.bre) → [ˈsoː.βɾɨ]

This suggests the following constraints in interaction: **FTBIN; TROCHEE; ALIGN-R(Ft, PrWd); ALIGN-R(HdFt, PrWd)**.

4. Marked metrification

If the **final vowel of an MMM** serves as the **nucleus of the penultimate syllable of the PrWd**, the same iterative formation of trochaic binary feet proceeds **from that penultimate syllable — rather than from the final syllable** — with the rightmost foot serving as the head foot and bearing primary stress; the final syllable is thus excluded from metrification and remains unstressed:

*acadé**mico*** /akadɛm-**ik**-o/ → a.ka.dɛ.m**i**.**ko** → (ˌa.ka)(ˈdɛ.m**i**)**ko** → [ˌɐ.kɐˈðɛː.mi.ku]
*agradá**vel*** /a-grad-a-**vel**-0/ → a.gra.da.**ve**.**l̥** → (ˌa.gra)(ˈda.**ve**)**l̥** → [ˌɐ.ɣɾɐˈðaː.vɛɫ]
número **/numɛr**-o/ → **nu**.**mɛ**.**r**o → (ˈ**nu**.**mɛ**)**r**o → [ˈnuː.mɨ.ɾu]
túnel /tunel-0/ → **tu**.**ne**.**l̥** → (ˈ**tu**.**ne**)**l̥** → [ˈtuː.nɛɫ]
sótão **/sɔtan**-o/ → **sɔ**.**tan**.o → (ˈ**sɔ**.**tan**)o → [ˈsɔː.tɐ̃ṵ̃]

*trabalhá**va**mos* /trabaʎ-a-**va**-mos/ → tra.ba.ʎa.**va**.mos → (ˌtra.ba)(ˈʎa.**va**)mos
→ [ˌtrɐ.βɐˈʎaː.vɐ.muʃ]
*trabalhá**ve**is* /trabaʎ-a-**va**-des/ → tra.ba.ʎa.**va**.des → (ˌtra.ba)(ˈʎa.**va**)des → [ˌtrɐ.βɐˈʎaː.vɐi̯ʃ]
*trabalha**r**des* /trabaʎ-a-**r0**-des/ → tra.ba.ʎa.**r̩**.des → (ˌtra.ba)(ˈʎa.**r̩**)des → [ˌtrɐ.βɐˈʎaɾ.muʃ]

By contrast, if the **final vowel of an MMM** does **not function as the nucleus of the penultimate syllable of the PrWd**, **unmarked metrification**, as described in Section 3, emerges:

*atoni**ci**dade* /a-tɔn-**ik**-i-dad-e/ → a.tɔ.n**i.s**i.da.de → (ˌa.tɔ)(ˌn**i.s**i)(ˈda.de) → [ˌɐ.tuˌni.siˈðaː.ðɨ]
***numer**oso* **/numɛr**-oz-o/ → **nu.mɛ.r**o.zo → (ˌ**nu.mɛ**)(ˈ**r**o.zo) → [ˌnu.mɨˈɾoː.zu]
*trabalha**va*** /trabaʎ-a-**va/** → tra.ba.ʎa.**va** → (ˌtra.ba)(ˈʎa.**va**) → [ˌtrɐ.βɐˈʎaː.vɐ]
*trabalha**r*** /trabaʎ-a-**r0/** → tra.ba.ʎa.**r̩** → (ˌtra.ba)(ˈʎa.**r̩**) → [ˌtrɐ.βɐˈʎaːɾ]

When a PrWd contains more than one MMM, metrification proceeds from the final syllable of the MMM whose final vowel function as the nucleus of the penultimate syllable of the PrWd:

***numéric**o* **/numɛr-ik**-o/ → **nu.mɛ.ri.k**o → **nu**(ˈ**mɛ.ri**)**k**o → [nuˈmɛː.ɾi.ku]

All of the above suggests the ranking: **ALIGN-R(Ft, σ_{final}_MMM) ≫ ALIGN-R(Ft, PrWd)**. Note that whereas syllabification refers also to morphological information, the metrification of syllable strings refers only to phonological information.

5. Stress clash avoidance (provisional)

When the metrification domain contains an **odd** number of syllables, to avoid a stress clash between a potential **word-initial unary foot** and the following **binary foot**, metrification is assumed to proceed as follows. These generalisations are **provisional** and call for further investigation.

5.1 Five or more syllables: only the leftmost foot is ternary

Only the **leftmost** foot is **ternary**; **subsequent** feet remain **binary** trochees:

papelaria /papel-ari-a/ → pa.pe.la.ri.a → (ˌ**pa.pe.la**)(ˈri.a) → [ˌpɐ.pɨ.lɐˈðaː.ðɨ]
computador /konput-a-dor-0/ → kon.pu.ta.do.r̩ → (ˌ**kon.pu.ta**)(ˈdo.r̩) → [ˌkõ.pu.tɐˈðoːɾ]
entendimento /entend-i-ment-o/ → en.ten.di.men.to → (ˌ**en.ten.di**)(ˈmẽ.to) → [ˌẽ.tẽ.diˈẽː.tu]

The constraint observations and violations of the winner and the losers are as follows:

Winner #(ˌσσσ)…#: NO *CLASH violation; **<u>NO PARSE(σ) violation</u>**; **ONE FTBIN violation**
Loser #σ(ˌσσ)…#: NO *CLASH violation; **ONE PARSE(σ) violation**; **<u>NO FTBIN violation</u>**
∴ **PARSE(σ) ≫ FTBIN**
Winner #(ˌσσσ)…#: **<u>NO *CLASH violation</u>**; NO PARSE(σ) violation; ONE FTBIN violation
Loser #(ˌσ)(ˌσσ)…#: **<u>ONE *CLASH violation</u>**; NO PARSE(σ) violation; ONE FTBIN violation
∴ ***CLASH** decides (tie-breaking).

This suggests the following constraints: **PARSE(σ) ≫ FTBIN**; ***CLASH**

5.2 Initial L-H exception: initial light syllable unfooted

If a **light initial syllable is immediately followed by a heavy syllable**, it remains **unfooted**:

levantamento /levant-a-ment-o/ → **le**.van.ta.men.to → **le**(ˌvan.ta)(ˈmen.to) → [lɨˌvɐ̃.taˈmẽː.tu]

Note that optional unary footing of the initial light syllable is not excluded, as in *levantamento* /levant-a-ment-o/ → **le**.van.ta.men.to → (ˌ**le**)(ˌvan.ta)(ˈmen.to) → [ˌlɨˌvɐ̃.taˈmẽː.tu].
The constraint observations and violations are as follows:

Winner # L (ˌH.σ)…#:
NO *CLASH violation; **ONE PARSE(σ) violation**; NO FTBIN violation; **<u>NO #WSP violation</u>**
Loser # (ˌL.H.σ)…#:
NO *CLASH violation; <u>NO PARSE(σ) violation</u>; ONE FTBIN violation; **ONE #WSP violation**
∴ **#WSP ≫ PARSE(σ)** — since PARSE(σ) ≫ FTBIN, as inferred in Section 5.1, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

Winner #L (ˌH.σ)…#:
<u>NO *CLASH violation</u>; **ONE PARSE(σ) violation**; NO FTBIN violation; NO #WSP violation
Loser #(ˌL)(ˌH.σ)…#:
ONE *CLASH violation; <u>NO PARSE(σ) violation</u>; ONE FTBIN violation; NO #WSP violation
∴ ***CLASH ≫ PARSE(σ)** — since PARSE(σ) ≫ FTBIN, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

This suggests the following constraint ranking: ***CLASH, #WSP ≫ PARSE(σ) ≫ FTBIN**

5.3 Three syllables: initial syllable unfooted

The **initial** syllable is **not footed**:

pessoa /peso-a/ → **pe**.so.a → **pe**(ˈso.a) → [**pɨ**ˈsoː.ɐ]
doutores /dout-or-0-s/ → **dou̯**.to.r0s → **dou̯**(ˈto.r0s) → [**do**ˈtoː.ɾɨʃ]
pequeno /peken-o/ → **pe**.ke.no → **pe**(ˈke.no) → [**pɨ**ˈkeː.nu]
levar /lɛv-a-r0/ → **lɛ**.va.ɽ → **lɛ**(ˈva.ɽ) → [**lɨ**ˈvaːɾ]
até /atɛ0/ → **a**.tɛ.0 → **a**(ˈtɛ.0) → [**ɐ**ˈtɛː]
após /apɔs0/ → **a**.pɔ.s̩ → **a**(ˈpɔ.s̩) → [**ɐ**ˈpɔːʃ]

Note that optional unary footing of the initial syllable, as in *levar* /lɛv-a-r0/ → lɛ.va.ɽ → (lɛ)(ˈva.ɽ) → [ˌlɨˈvaːɾ], is not excluded. The constraint observations and violations of the winner candidate and the loser candidates are as follows:

Winner #σ(ˈσσ)#: **NO *CLASH violation**; **ONE PARSE(σ) violation**; NO FTBIN violation
Loser #(ˌσ)(ˈσσ)#: **ONE *CLASH violation**; **NO PARSE(σ) violation**; ONE FTBIN violation
∴ ***CLASH ≫ PARSE(σ)** — since PARSE(σ) ≫ FTBIN, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

This also suggests the following constraint ranking: ***CLASH, #WSP ≫ PARSE(σ) ≫ FTBIN**

6. The constraint set (provisional)

From the discussion in Sections 3, 4 and 5, the following set of constraints and their rankings are inferred:

***CLASH, #WSP ≫ PARSE(σ) ≫ FTBIN;**
TROCHEE;
ALIGN-R(Ft, σ_{final}_MMM) ≫ ALIGN-R(Ft, PrWd);
ALIGN-R(HdFt, PrWd)

7. Orthographic corollary

When **unmarked metrification** is triggered by an MMM, the stressed nucleus is **typically indicated orthographically** by an acute accent or a circumflex accent, since the outcome **deviates from the default stress position** — e.g. *âmbar*, *côdea*, *número*, *pêssego*, *sótão*, *túnel*, *académico*, *agradável*, *trabalhávamos*, *trabalháveis*. This close correlation suggests that **speakers' prosodic intuitions** about **stress location** are reflected in the standard orthography, thereby providing independent support for the analysis.

8. Conclusion

Thus, Portuguese stress can be entirely reduced to **qualityless vowel /0/ with one mora +** ***metrically marked morphemes*** **+ general metrical theory**, eliminating the need for morphologically-driven stress assignment.

Reference

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